By substituting this inequality, we get an upper bound on our equation,

$$\mathrm{Var}(\mathcal{A}(h,a)) \leq \mathrm{Var}(\mathcal{A}_i(h,a)) + \mathrm{Var}(\mathcal{A}_{\neg i}(h,a)) + 2\sqrt{\mathrm{Var}(\mathcal{A}_i(h,a))\mathrm{Var}(\mathcal{A}_{\neg i}(h,a))}$$
$$\mathrm{Var}(\mathcal{A}(h,a)) \leq (\sqrt{\mathrm{Var}(\mathcal{A}_i(h,a))} + \sqrt{\mathrm{Var}(\mathcal{A}_{\neg i}(h,a))})^2$$

The above equation shows that the variance of the policy gradient update grows approximately linearly with the number of agents in the multi-agent system. This increase in variance reduces the signal-to-noise ratio of the policy gradient, necessitating more updates for effective learning. Proper credit assignment can mitigate this issue by enhancing the signal-to-noise ratio, thereby facilitating more sample-efficient learning.

4 Method

4.1 Definition of reward redistribution function

In this paper, we address the challenge of temporal and agent credit assignment in fully cooperative multi-agent systems with episodic global rewards. Our goal is to learn a reward redistribution function that preserves the optimal policy of the original reward function of the environment. We aim to achieve this by defining a reward redistribution function that decomposes the episodic trajectory reward, also known as trajectory return, to each agent based on their contribution to the team's outcome at every time step.

We premise that the joint history and action at the final timestep of the multi-agent trajectory serve as a good proxy for predicting the episodic global reward. Consequently, we predict the per timestep reward for each agent by assessing the contribution of its state-action tuple towards generating this joint history of the multi-agent system.

Assumption 1. *The reward redistribution function $r_{i,t}$, which assigns the reward received by agent i at time step t, is determined by analyzing the importance of each state-action tuple against the joint history and action at the final timestep that predicts the episodic global reward.*

$$r_{global,episodic}(\tau) = \sum_{i=1}^{N}\sum_{t=1}^{|\tau|} r_{i,t}(h_{i,t}, a_{i,t}, h_{\neg i,t}, a_{\neg i,t}, h_{|\tau|}, a_{|\tau}), \tag{1}$$

This assumption aligns with the framework adopted by prior works (Xiao et al., 2022; Ren et al., 2021; Efroni et al., 2021) which also assumes that the episodic return has some structure in nature, e.g., a sum-decomposable form.

4.2 Assembling the reward function

We propose a method to redistribute the trajectory returns temporally, assigning credit to each time step in the multi-agent trajectory. Subsequently, the temporally redistributed rewards are further decomposed across agents based on their individual contributions, ensuring that the relationship expressed in (1) is maintained.

Given the dual nature of credit assignment—attributing relative credit to: (1) each time step in the multi-agent trajectory, and (2) each agent at every time step—we can derive the relationship between the trajectory return, $r_{\text{global,episodic}}$, and the redistributed reward received by agent i at time step t, $r_{i,t}$.

To formalize, we assume the following process:

Assumption 2. *The trajectory return $r_{global,episodic}$ is redistributed temporally to obtain rewards for each time step, $r_{global,t}$, such that:*

$$r_{global,episodic}(\tau) = \sum_{t=1}^{|\tau|} r_{global,t}(h_t, a_t, h_{|\tau|}, a_{|\tau|}).$$

Subsequently, these temporally redistributed rewards are decomposed across agents:

$$r_{global,t}(h_t, a_t) = \sum_{i=1}^{N} r_{i,t}(h_{i,t}, a_{i,t}, h_{\neg i,t}, a_{\neg i,t}, h_{|\tau|}, a_{|\tau|}),$$

ensuring that:

$$r_{global,episodic}(\tau) = \sum_{i=1}^{N} \sum_{t=1}^{|\tau|} r_{i,t}(h_{i,t}, a_{i,t}, h_{\neg i,t}, a_{\neg i,t}, h_{|\tau|}, a_{|\tau|}).$$

This two-step decomposition process—first temporally and then across agents—ensures that each agent's contribution at each time step is appropriately credited, thereby providing a robust framework for reward redistribution in multi-agent systems.

Let's define a function $w_t \sim W_\omega(h_t, a_t, h_{|\tau|}, a_{|\tau|}) : (\mathcal{H} \times \mathcal{A}) \times (\mathcal{H}_{|\tau|} \times \mathcal{A}_{|\tau|}) \to \mathbb{R}$ that redistributes the rewards across the temporal axis of the multi-agent trajectory. Thus, we can express the multi-agent temporal reward at an arbitrary time step t as

$$r_{\text{global},t} = w_t r_{\text{global,episodic}}(\tau)$$

Similarly, let's define a function $w'_{i,t} \sim W_\kappa(h_{i,t}, a_{i,t}, h_{\neg i,t}, a_{\neg i,t}) : \mathcal{H}_i \times \mathcal{A}_i \times \mathcal{H}_{\neg i} \times \mathcal{A}_{\neg i} \to \mathbb{R}$ that redistributes the temporal rewards at an arbitrary time step t across agents. Hence, now we can express the reward that agent i receives as

$$r_{i,t} = w'_{i,t} r_{global,t}$$

Finally, deriving the relationship between $r_{i,t}$ and $r_{\text{global,episodic}}(\tau)$ for an arbitrary time-step t

$$r_{i,t} = w'_{t,i} w_t r_{\text{global,episodic}}(\tau)$$

Based on the definition of reward redistribution function

$$\begin{aligned}
\sum_{i=1}^{N} \sum_{t=1}^{|\tau|} r_{i,t}(h_{i,t}, a_{i,t}, h_{\neg i,t}, a_{\neg i,t}, h_{|\tau|}, a_{|\tau|}) &= r_{\text{global,episodic}}(\tau) \\
(\sum_{i=1}^{N} \sum_{t=1}^{|\tau|} w'_{t,i} w_t r_{\text{global,episodic}}(\tau) &= r_{\text{global,episodic}}(\tau) \\
\sum_{i=1}^{N} \sum_{t=1}^{|\tau|} w'_{t,i} w_t &= 1 \\
\sum_{t=1}^{|\tau|} (\sum_{i=1}^{N} w'_{t,i}) \times w_t &= 1
\end{aligned}$$

The solution for the above equation is,

$$\sum_{i=1}^{N} w'_{t,i} = 1 \tag{2}$$

$$\sum_{t=1}^{|\tau|} w_t = 1 \tag{3}$$

To construct the new reward function $\mathcal{R}_{\omega,\kappa}$, we incorporate the original reward $\mathcal{R}_\zeta$ and the redistributed credit $r_{i,t}$ that each agent i receives at time step t. This redistributed credit reflects the agent's relevance to the final outcome of the multi-agent system. The relevance of each state-action tuple is determined by the reward redistribution functions W_ω and W_κ, which effectively assign the trajectory rewards temporally and across agents.

$$\begin{aligned} R_{\omega,\kappa}(s_t, a_t, s_{t+1}) &= R_\zeta(s_t, a_t, s_{t+1}) + r_{i,t} \\ R_{\omega,\kappa}(s_t, a_t, s_{t+1}) &= R_\zeta(s_t, a_t, s_{t+1}) + w_{t,i} w_t r_{\text{global,episodic}}(\tau) \end{aligned} \tag{4}$$

This formalizes the reward redistribution process, ensuring that each agent receives a reward proportionate to its contribution to the overall team performance at each timestep.

4.3 Optimal Policy Preservation

To ensure that the optimal policy learned using the densified reward function is also optimal in the environment's original reward function, we establish the following theorem:

Theorem 2. *Let's consider two Dec-POMDPs as defined in subsection 3.1, $\mathcal{M}_{env} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{T}, \mathcal{O}, \mathcal{N}, \mathcal{R}_\zeta, \rho_0, \gamma)$ and $\mathcal{M}_{rrf} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{T}, \mathcal{O}, \mathcal{N}, \mathcal{R}_{\omega,\kappa}, \rho_0, \gamma)$. The only distinction between $\mathcal{M}_{env}$ and $\mathcal{M}_{rrf}$ are the reward functions. If π^*_θ is the optimal policy in $\mathcal{M}_{rrf}$ then π^*_θ is also optimal in $\mathcal{M}_{env}$.*

Proof. We know that π^*_{theta} is optimal in $\mathcal{M}_{\text{rrf}}$. For π^*_{theta} to be optimal in $\mathcal{M}_{\text{env}}$, we need to show that $\mathcal{R}_{\omega,\kappa} = \mathcal{R}_\zeta + \mathcal{F}(s_t, a_t, s_{t+1})$ where $\mathcal{F}(s_t, a_t, s_{t+1})$ is a potential based shaping function which is a necessary and sufficient condition for optimal policy preservation 3.3.

It is therefore sufficient to show that the equation (4) takes the form $\mathcal{R}_{\omega,\kappa}(s_t, a_t, s_{t+1}) = \mathcal{R}_\zeta(s_t, a_t, s_{t+1}) + \gamma\phi(s_{t+1}) - \phi(s_t)$. Comparing this format to equation (4), assuming $\gamma = 1$ we arrive at $\phi(s_{t+1}) - \phi(s_t) = w'_{t,i} w_t r_{\text{global,episodic}}(\tau)$. This relation holds for $\phi(s_t) = r_{\text{global,episodic}}(\tau)(\sum_{t'=0}^{t} w'_{t',i} w'_t)$ □

This result ensures that if an arbitrary policy π_θ when trained using the reward function $\mathcal{R}_{\omega,\kappa}$ in $\mathcal{M}_{\text{rrf}}$ converges to an optimal policy π^*_θ then π^*_θ will also be optimal for the original reward function $\mathcal{R}_\zeta$ in $\mathcal{M}_{\text{env}}$.

4.4 Policy Gradient Update Equivalence with Reward Redistribution

In this subsection, we establish that the policy gradient update for an arbitrary agent k, derived from the reward redistribution function, shares the same direction but exhibits a smaller magnitude than the policy gradient update in the environment's original reward function. Furthermore, this ensures that the policy update trajectory towards the optimal policy for an arbitrary initial policy is preserved for every agent.

Proposition 1. *Let π_θ be the parametric policy in a decentralized execution paradigm, where the joint policy is expressed as a product of individual agent policies $\pi_\theta = \prod_{k=1}^{N} \pi_{\theta_k}$ (Oliehoek & Amato, 2016; Amato, 2024). The policy gradient update for an arbitrary agent k under the reward redistribution function $R_{\omega,\kappa}$ follows the same direction as the policy gradient update under the environment's original reward function R_ζ, preserving the policy update trajectory towards the individual optimal policies.*

Proof. Consider the policy gradient update for agent k under the reward redistribution function:

$$\nabla_{\theta_k} \mathbb{E}_{\pi_{\theta_k}} \left[\sum_{t=1}^{|\tau|} r_{k,t} \right] = \nabla_{\theta_k} \mathbb{E}_{\pi_{\theta_k}} \left[\delta(\tau) r_{\text{global,episodic}}(\tau) \right] = \nabla_{\theta_k} \mathbb{E}_{\pi_{\theta_k}} \left[\sum_{i=1}^{N} \sum_{t=1}^{T} \delta_t r_{i,t} \right],$$